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The socioeconomic returns to citizenship: A randomized controlled trial

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Based on observational studies, conventional wisdom suggests that citizenship carries economic benefits. We leverage a randomized experiment from New York where low-income registrants with permanent residency who wanted to become citizens entered a lottery to receive fee vouchers to naturalize. Voucher recipients were about 36 p.p. more likely to naturalize. Yet, we find no discernible effects of access to citizenship on multiple economic outcomes, including income, credit scores, access to credit, financial distress, and employment. Leveraging a multidimensional immigrant integration index, we similarly find no measurable effects on noneconomic integration. However, we do find that citizenship reduces fears of deportation. Explaining divergence from past studies, our results also reveal evidence of positive selection into citizenship, suggesting that observational studies are susceptible to selection bias.

citizenship | naturalization | immigrant | RCT | integration

Effective integration of immigrants is a long-standing policy challenge for governments throughout the world (1, 2). Immigrants often face systematic disadvantages in host country labor markets due to discrimination, lack of recognition of foreign credentials, limited language proficiency, and other barriers (3–9). Recognizing these disadvantages, policy makers and researchers have proposed easing the route from permanent residency to citizenship as a potential catalyst for economic integration.

While there have been a few studies with mixed findings, the overwhelming consensus in the academic literature is that citizenship has positive effects over permanent residency across a range of outcomes (see *SI Appendix*, Table S25 accompanied with a quantitative accounting of the percentage of studies reporting positive effects on the acquisition of citizenship). Illustrative of this, a comprehensive expert report by the National Academies of Sciences concluded that "U.S. citizenship [...] improves employment outcomes, wage growth, and access to better jobs." (10, p. 165). There have been dozens of observational studies reporting significant economic returns to citizenship on earnings, including a 6 to 14% citizenship premium for the United States (e.g., refs. 11–13) and mostly positive effects across many immigrant-accepting countries with magnitudes ranging up to 28% in France (14). With a few exceptions that employ regression discontinuity designs (e.g. refs. 15–17), research in this area has been hampered by the difficulty of accounting for unobserved differences—such as motivation and perseverance—between naturalized and nonnaturalized immigrants, which are likely correlated with economic outcomes. To the best of our knowledge, random assignment of citizenship has not been utilized to date, limiting the ability to infer causal relationships.

Between 2016 and 2018, New York State implemented a program to help low-income immigrants to naturalize. Eligible lawful permanent residents could sign up for a lottery that randomly selected some registrants to receive a voucher to pay for the naturalization application fee. This lottery provides a unique opportunity to study the impacts of access to citizenship using a randomized control design. In addition, the lottery helped the implementing agency to allocate vouchers without bias among those eligible as demand for vouchers exceeded supply. Furthermore, it allows us to identify the returns to citizenship in a population that the literature has identified as the most likely to gain—immigrants that are 1) eligible and motivated to naturalize and 2) in the lower part of the income distribution.

We focus on four research questions. First, what are the causal effects of access to citizenship on immigrants' financial outcomes, including income, credit scores, financial distress, and access to credit? Second, if there are any economic returns, are they driven by improvements in employment, labor force participation, human capital investments, or occupational upgrading? Third, what are the impacts of access to citizenship on

Significance

Citizenship is widely believed to improve immigrants' economic and social integration, but existing evidence relies on observational studies prone to selection bias. This study presents a randomized controlled trial to assess the causal effects of citizenship. In a naturalization campaign in New York City, low-income immigrants were randomly offered fee vouchers to apply for citizenship. Although vouchers significantly increased naturalization, we find no measurable impact on economic outcomes, financial well-being, or broader integration indicators. The only clear benefit was a reduction in deportation fears. These findings challenge prevailing assumptions about the transformative power of citizenship over permanent residency and highlight the importance of rigorous causal evidence in shaping immigrant integration policy.

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noneconomic outcomes including the political, social, psychological, linguistic, and navigational integration of immigrants and their fears of deportation? Fourth, is there selection bias that impedes the identification of a causal effect of citizenship?

Our analysis combines individual-level data from the program registration, follow-up surveys of registrants, and matched administrative records from a credit bureau containing financial information in the years prior to and for up to five years following the voucher lottery. We compare the outcomes of the treatment group of registrants who were selected to receive the naturalization fee voucher to those of the control group who were not selected for the voucher to estimate intention-to-treat (ITT) effects. We also estimate the local average treatment effects (LATE) of citizenship, using the voucher lottery as an instrument for self-reported naturalization. Unless indicated otherwise, all our analyses follow a preanalysis plan that we registered before linking the outcomes with the treatment indicators.

Our study yields several findings. First, consistent with previous findings (18), we find that being selected to receive a fee voucher in the lottery resulted in a large increase in naturalization on the order of 36 percentage points (first-stage partial F-statistic $>$ 200). However, this did not translate into discernible improvements in financial outcomes measured in matched credit records. In particular, the estimates suggest that for up to five years after the lottery, average income among registrants in the treatment group was similar to that in the control group. We also find no discernible effects on credit scores, various measures of financial distress, and measures of access to credit. These null findings are stable across subgroups and across quantiles of the outcome distributions. We similarly find no discernible effects when we replicate the models using survey-based measures of income and financial distress. In addition, we examine whether the impacts grow over time but find no discernible evidence of a change in slopes between the treatment and control group.

Our null findings are meaningfully precise. The ITT effect on income from our panel model is centered at 0.2%, equivalent to a \$102 gain, with a 95% CI ranging from a \$767 decrease to a \$1,022 increase. Estimates for LATE and other financial outcomes-such as credit score, financial distress, and credit access-are also relatively precise.

Second, consistent with these null findings, we find no support for envisioned mechanisms that explain how citizenship can improve economic integration (e.g., refs. 11 and 19). In particular, using data on educational loans and self-reported educational enrollment, we find no discernible evidence that access to citizenship encourages immigrants to invest in host-country specific human capital. Similarly, we find no measurable effects on self-reported employment, labor force participation, and occupational upgrading.

Third, the null findings on economic returns raise the question of whether citizenship may improve other (noneconomic) dimensions of integration, such as immigrants' English skills, political engagement, their social ties with natives, or psychological attachments to the United States. To examine this question, we leverage a survey-based multidimensional measure of immigrant integration (20). As with the economic returns, we find no support for the idea that access to citizenship resulted in greater political, social, psychological, navigational or linguistic integration. However, we do find that citizenship reduced fears of deportation.

Fourth, our study also reveals the presence of positive selection effects. When we use the nonexperimental variation and compare naturalizers with nonnaturalizers, we find that citizenship is associated with higher income, fewer third-party debt collections, and a higher probability of having an open credit line. This finding suggests that absent random assignment, the selection into citizenship is an obstacle to causal identification even with panel data.

Fifth, combining the experimental data with population-level data from all low-income immigrants in New York and other metro areas, we find that the population average treatment effects match those in the experimental sample. These findings support the external validity of the study for the US context.

This study contributes to the literatures on economic integration of immigrants and the design of citizenship policies.* Econometric tests of the citizenship premium have been conducted in at least eleven OECD countries.† These studies have given rise to a conventional wisdom that the citizenship premium over permanent residency is significant, especially for immigrants from poorer countries and those that are more disadvantaged in the labor market. Yet, prior tests have relied exclusively on observational studies and our study adds to this literature providing experimental estimates of the effects of citizenship. Our findings run counter to this literature and temper optimism about the promise of citizenship over permanent residency as an important policy lever to improve the socioeconomic integration of immigrants.

1. Materials and Methods

1.1. Experimental Design. Our study draws on a statewide program in New York that promoted naturalization among low-income legal permanent residents. Operating annually from 2016 to 2018, the initiative launched multilingual outreach campaigns to encourage eligible immigrants to register-via online, phone, or in-person options-for support in applying for U.S. citizenship. Eligibility required being 18 or older, residing in New York State, satisfying federal naturalization criteria, and having household income at or below 300% of the Federal Poverty Guidelines (FPG).

Registrants with incomes below 150% of the FPG or receiving means-tested public benefits were informed that they might qualify for an existing federal fee waiver. Those with incomes between 150% and 300% of the FPG, and not receiving benefits, were entered into a lottery to receive a voucher covering the full naturalization application fee (ranging from \$680 to \$725, depending on the year).

The voucher lottery was conducted separately in 2016, 2017, and 2018. After each annual registration window closed, all registrants eligible for the voucher lottery were randomly assigned to one of two groups: The treatment group received a voucher covering the full application fee, while the control group received no voucher. Vouchers were administered through designated nonprofit immigrant service providers, paid directly to the United States Citizenship and Immigration Services (USCIS), and could be used only for the purpose of applying for naturalization.

To facilitate access to services, the lottery was stratified by geographic "blocks" constructed using registrants' addresses and expected application fees (full or reduced). In each year, lotteries were conducted only in blocks with more eligible registrants than available vouchers. These conditions were met only in parts of New York City and Long Island, so the analysis is limited to those areas.

Overall, 2,802 registrants were randomized across the three years, with 1,442 assigned to treatment and 1,360 to control. The cohort sizes by registration year

*To examine the conventional wisdom in the literature, we accumulated a systematic sample of fifty-two studies estimating the citizenship premium for a broad set of outcomes from wages to health. We first chose highly cited papers in Google Scholar that cited one of two seminal papers on citizenship effects (11, 12). We then collected a snowball sample of articles based on keywords such as immigration and citizenship. Researchers of 69% of these papers inferred from their data that there was a significant earnings premium for at least one country under study and/or a major immigrant group. Three literature reviews reported on average positive returns to acquiring citizenship. A full list of these studies is in *SI Appendix*, Table S25.

†These studies include the following: the United States (3, 13, 21–29); Canada (12, 26, 30–33), Germany (34–39); Sweden (32, 40–44); Norway (45, 46); Switzerland (17, 34, 47); Netherlands (36, 48–50); France (14, 51, 52), Austria (43); Denmark (44); Belgium (53); and other countries as well in a variety of cross-national comparisons (13, 38, 53, 54).

are as follows: 836 in 2016 (309 treated), 746 in 2017 (490 treated), and 1,220 in 2018 (643 treated). Additional details on the study design, data, outcomes, and sample are provided in *SI Appendix*. This research was approved by the Stanford University IRB protocol #34554.

1.2. Data.

1.2.1. Sources. We use three primary data sources. First, the program's registration data includes pretreatment demographic and socioeconomic characteristics of eligible registrants, such as age, gender, education, employment, income, and green card receipt year. It also indicates registration year, randomization block, and voucher assignment status.

Second, we use follow-up survey data, collected annually for each cohort starting roughly one year after voucher assignment. The surveys measured naturalization status (application submitted and citizenship attained) and a broad set of integration outcomes. For all three cohorts (2016–2018), we observe outcomes for up to three years postlottery, with four- and five-year data available for the 2017 and 2016 cohorts, respectively. Our main analyses focus on the three-year outcomes, but we also report robustness checks using two-year and longer-term data.

Third, we match registrants to anonymized credit bureau records using name, address, and birth date. Registrants consented to data linkage for research purposes. The algorithm yielded a 93.3% match rate. To ensure anonymity, the bureau returned only deidentified records, including only outcomes and coarsened covariates indicators for household income (above/below median) and language of registration (English/other).

The credit data provides annual snapshots of financial outcomes–income, credit scores, credit access, and financial distress-from 2016 to 2021. These allow us to measure pre- and posttreatment outcome dynamics. The number of preintervention periods varies by cohort: one for the 2016 cohort, two for 2017, and three for 2018. Posttreatment, we observe five years for 2016, four for 2017, and three for 2018. This structure enables us to assess baseline balance and estimate short- and medium-term effects across cohorts.

1.2.2. Outcomes. The outcome variables are measured in the administrative credit bureau data and the follow-up surveys. The credit data provides objective financial measures not subject to self-report bias, while surveys capture broader indicators of integration. We examined household income (log-transformed), credit scores (VantageScore v3.0), indicators of financial distress (e.g., delinquencies and collections), and access to credit (e.g., having a credit score, having a "thick file" (three or more open lines of credit), open credit lines, and total revolving credit). We also include a binary indicator of student loan activity as a proxy for educational investment.

The surveys allow us to assess additional economic and noneconomic outcomes. Economic indicators include self-reported household and personal income (log-transformed), employment, labor force participation, and ability to handle unexpected expenses. We also measure occupational upgrading using SOC-coded job titles linked to ACS-based wage and education ranks. Noneconomic integration is assessed using the IPL-12 multidimensional index, which covers six dimensions: economic, political, social, psychological, navigational, and linguistic. Finally, we include a measure of perceived deportation risk, based on responses to a four-point Likert question on worry about deportation.

1.2.3. Preregistration. This study was preregistered (AEARCTR-0006790, AEARCTR-0007148) prior to linking treatment and outcome data. Most analyses follow the preanalysis plan, with three exceptions: 1) we report three-year (instead of two-year) outcomes due to extended data collection, though two-year results are similar and shown in *SI Appendix*; 2) a deportation fear question was added in the final survey wave, inspired by ref. 55; and 3) a second credit score outcome was dropped due to credit bureau restrictions.

1.3. Statistical Methods. We use canonical, prespecified methods for analysis of randomized encouragement designs to measure the impacts of the intervention (56–59).

1.3.1. Three-year outcomes model. For our baseline models, we estimate the ITT effects using the following equation:

$$y_i = \alpha_0 + \alpha_1 \mathrm{LOTTERY}_i + \theta X_i + B_k + \epsilon_i, \qquad [1]$$

where y_i is an outcome variable measured three years after the intervention for each cohort; $LOTTERY_i$ is a dummy variable for whether or not registrant i won in the lottery and was offered a voucher; the term X_i is a vector of prespecified, prerandomization control variables; B_k is a vector of dummy variables that indicate the geographic lottery randomization blocks, and ϵ_i is the error term. When using the credit bureau data X_i includes indicators for above/below median household (HH) income and English language ability and the pretreatment outcome. When using the survey data X_i includes income, age, female, origin dummies (Dominican Republic, Ecuador and Colombia), year green card was received, education level dummies, language dummies (English and Spanish), and time between registration and voucher lottery. We use robust SEs and employ block-level inverse probability weights to account for the unequal probability of treatment assignment within each block. We do not cluster the SEs because the randomization occurred at the individual level (60).

The coefficient of interest is α_1 and it identifies the ITT effect of winning the voucher in the lottery. It can be interpreted as the impact of obtaining access to citizenship by overcoming the financial hurdle of paying for the naturalization fee. The ITT effect is causally identified by virtue of the randomized lottery.

Next, to obtain the LATE of citizenship for compliers, we estimate the following equation using two-stage least squares (2SLS):

$$y_i = \beta_0 + \beta_1 \mathrm{CITIZENSHIP}_i + \phi X_i + B_k + \epsilon_i, \qquad [2]$$

where $CITIZENSHIP_i$ is a binary treatment variable for whether or not a registrant reported having submitted their citizenship application during the first check-in survey. In this equation, $CITIZENSHIP_i$ is instrumented by $LOTTERY_i$ to accommodate noncompliance and address selection bias. All other aspects of the estimation remain the same as in the ITT model.

The coefficient β_1 identifies the LATE of citizenship for compliers, i.e., registrants who would naturalize if given the fee voucher and would not otherwise. The causal interpretation of this parameter requires the additional identifying assumption that winning the voucher in the lottery has no effect, on average, on the outcomes of interest that does not operate via the impact of the voucher on naturalization. Given that the voucher was sent directly to USCIS and could only be used to pay for the naturalization application fee, we believe that this assumption is reasonable. Yet, the exclusion restriction may not strictly hold. One possible violation is that winning the lottery could, in theory, have a positive psychological effect, but it is not likely that this would have a lasting impact on the outcomes we study. Another possible violation is that for always-takers (i.e., registrants who would always naturalize regardless of winning the voucher), the voucher could act as a one-time substitute for cash. Given that the average registrant had a prelottery household income of about $46,630, the voucher represents a one-time 1.56% increase. It is theoretically possible that this could have resulted in a short-term positive indirect effect on some of the outcomes unrelated to naturalization. However, it seems unlikely that this effect would persist for multiple years after the lottery. This scenario, if present, might introduce a small upward bias in our LATE estimates.

Whether one prefers the ITT or the LATE is largely a matter of analytical interest and methodological considerations; both estimands are used in the literature on the effects of citizenship. The ITT captures the impact of access to citizenship and is arguably more relevant from a policy perspective. Policy makers can change eligibility criteria or design encouragements for immigrants to apply for naturalization, but cannot typically force them to naturalize. The ITT has therefore been the focus in many recent studies of citizenship (e.g., refs. 13, 15–17, 37, and 61). The LATE captures the effect of citizenship per se for compliers, but relies on the exclusion restriction assumption.

1.3.2. Panel model. Here, we analyze models that leverage the full six-year panel structure of the credit bureau data across the three registration cohorts. We also employ this model in a six-year panel when using the survey data. In particular, we estimate the following panel equation:

$$y_{it} = \delta_i + \sigma_t + \gamma LOTTERY_{it} + \epsilon_{it}, \qquad [3]$$

where the terms δ_i and σ_t represent individual and year fixed effects, and $LOTTERY_{it}$ is the randomized treatment indicator coded as one for observations from years when a registrant was assigned to the voucher and zero otherwise. The coefficient of interest is γ which identifies the ITT effect pooling together the short- and medium-term effects of the intervention.

For the LATE analysis, we estimate the equivalent 2SLS model where $CITIZENSHIP_{it}$ is instrumented by $LOTTERY_{it}$. When using the credit bureau data, we focus on a sample of individuals who are successfully matched by the credit bureau in each year during 2016–2021 (i.e., balanced six-year panel). When using the survey data, these models are limited to outcome variables which were measured prior to the randomization. SEs are clustered at the individual level and we weight each observation by the block-level inverse probability of being randomized into the treatment group. In *SI appendix*, we also show other panel models that allow for heterogeneous treatment effects by cohort and time.

1.3.3. Dynamic effects model. Following the analysis in ref. 11, we also estimate panel models where we allow the effects of access to citizenship to change over time. In particular, we estimate the following equation:

$$\begin{aligned} y_{it} = \delta_i + \sigma_t + \theta_1 LOTTERY_{it} + \\ + \theta_2 (LOTTERY_{it} \times YEARS\_SINCE_{it}) \\ + \theta_3 YEARS\_SINCE_{it} + \epsilon_{it}, \end{aligned} \quad [4]$$

where the term θ_1 identifies the effect of the intervention in the first year and θ_2 measures how the effect changes for each additional year following the intervention (as measured by the variable $YEARS\_SINCE_{it}$). In other words, θ_2 identifies the linear change in the outcome trends between treated and control registrants in the years following the intervention. All other aspects of the model remain as specified above.

1.3.4. Balance, match rates, and survey attrition. We conducted multiple checks to validate the experimental design (*SI Appendix*). First, balance tests confirm that treatment and control groups are statistically similar across pretreatment covariates. Second, credit bureau match rates are high (93.3%) and balanced across groups, with no significant differences in matching into one-year, four-year, or six-year panels. Third, while the overall response rate to the three-year survey was about 47%, regression analyses show no significant interaction between treatment status and baseline characteristics in predicting a response. Robustness checks using inverse probability weighting yield similar results, supporting the validity of our survey-based estimates. Overall, all these checks reinforce the conclusion that randomization was successful and that the treatment and control groups are otherwise identical.

1.3.5. Descriptive statistics. *SI Appendix*, Tables S1 and S2 report pretreatment summary statistics. The average age of registrants in our sample is 42.1 y, the sample contains slightly more women (54.4%) than men, and the most common origin country is the Dominican Republic (27.3%). Most registrants completed the registration in English (63%), followed by those who used Spanish (34%). Most registrants have an open credit line (88.0%) and a "thick file" (65.6%). Virtually everyone (96.7%) has a VantageScore with an average VantageScore of 629, which is generally considered "Fair," but it falls below the national average. Nearly four in five registrants (79.0%) are in the labor force and the vast majority of those (93.2%) are employed.

1.4. Results.

1.4.1. First stage. We begin by examining whether the random assignment of the vouchers via the lottery increased the probability of naturalization. Table. 1 presents the first-stage estimates from regressing the citizenship indicator on the voucher indicator and the indicators for the geographic randomization blocks. The first two columns show the effect of voucher assignment for registrants who are matched in the four-year panel of credit bureau data. The last two columns show the effect for respondents in the survey data.

Table 1. First stage results: Effect of voucher on citizenship

	Credit bureau data		Survey data	
Outcome	Citizenship		Citizenship	
Voucher	0.346***	0.354***	0.366***	0.369***
	(0.024)	(0.024)	(0.021)	(0.021)
Partial *F*-stat.	209.29	224.61	293.81	306.41
N	1,564	1,541	1,984	1,984
$\bar{Y}$	0.62	0.63	0.62	0.62
Controls	No	Yes	No	Yes

Each entry is an estimated coefficient from a separate regression. Robust SEs are shown in parenthesis. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$.

We find that the voucher assignment increased the likelihood of submitting a citizenship application by about 35 to 37 percentage points, on average (also see ref. 18). As expected, given the random lottery assignment, the estimate of the effect of the voucher on the increase in citizenship is nearly identical when we control for the baseline covariates in columns 2 and 4. The first stage partial *F*-statistics are above 200, indicating that the voucher instrument is strong. Overall, these results suggest that the application fees constitute a financial barrier to citizenship for low-income immigrants.

1.4.2. Treatment effect estimates: Credit bureau data.

1.4.2.1. Three-year outcomes. Before turning to the formal impact estimates from the regression models, we provide graphical evidence that illustrates the main findings. We construct "event study"-type plots of the mean outcome values for the treatment and control groups from two years prior up to five years after the voucher assignment. Year zero refers to the year of voucher lottery.

The results are presented in Fig. 1. We focus on the four main outcomes–(log) income (*Top Left* panel), the VantageScore (*Top Right*), the index that combines the measures of financial distress (*Bottom Left*), and the index that combines the access to credit measures (*Bottom Right*). Consistent with the random assignment of the vouchers, the treatment and control group have similar trends prior to the intervention across all four outcomes. Moreover, both groups also have very similar trends for up to five years following the intervention, demonstrating that naturalization vouchers had no discernible effect on any of the four outcomes.

Next, we turn to the formal effect estimates from the regression models. The *Top* panel in Table. 2 presents the ITT estimates from the baseline specification Eq. **1**. Consistent with the graphical evidence, we find that receipt of the naturalization voucher had no discernible positive impact on the financial outcomes of registrants. For income, the point estimate is 0.0% with a 95% CI that rules out effects below −2.4% and above 2.4%. Moreover, we find no discernible effects on the measures of financial distress or access to credit. If anything, we find a small drop in VantageScore scores and the probability of having an open credit line, but these effects are not robust across specifications and we therefore refrain from interpreting them as negative effects.

Our estimates are sufficiently precise to rule out substantively meaningful effects. The average household income is around \$58,606 three years after the lottery for this estimation sample, which implies that the point estimate of the income effect amounts to a change of \$0 and the 95% CI ranges from a decrease of about \$1,406 to an increase of about \$1,406. For the effect on the VantageScore, the point estimate is −12 points while the 95% CI ranges between −22 and −2 points which is rather small compared to a mean score of 655 points and a SD of 142. For the effect on our index of distress, the point estimate is 0.07 with a 95% CI that ranges between −0.02 points and above 0.15 points compared to a mean index score of 1.1 and a SD of 1.3. For the effect on our index of access to credit the point estimate is −0.01 with 95% CI ranges from −0.03 points to 0.01 points compared to a mean index score of 0.9 and a SD of 0.2.

The *Bottom* panel in Table. 2 presents the LATE estimates from the baseline specification Eq. **2**. Again, we find no discernible evidence that citizenship improved the financial outcomes of immigrants. As expected, the LATE estimates are about three times larger than the ITT estimates.

1.4.2.2. Panel. To leverage the full credit bureau panel data for the three registration cohorts over the entire 2016–2021 period, we now present estimates from a panel regression. Table. 3 presents the ITT and LATE estimates following equation Eq. **3**. The results mirror the null findings from the earlier models but are slightly more precise. For income, the ITT estimate is 0.2% with a 95% CI that rules out effects below −1.5% and above 2.0%. The corresponding LATE estimate is 1.8% (95% CI = −1.8%, 5.3%). The null findings are similar for the other financial outcomes including the measures of financial distress and access to credit. Again, the point estimate of the ITT effects on VantageScore is negative (−6.8 points decrease), but the 95% CI is consistent with no effect (95% CI = −15.8, 2.3). The same is true for the effect on the probability of having an open credit line.

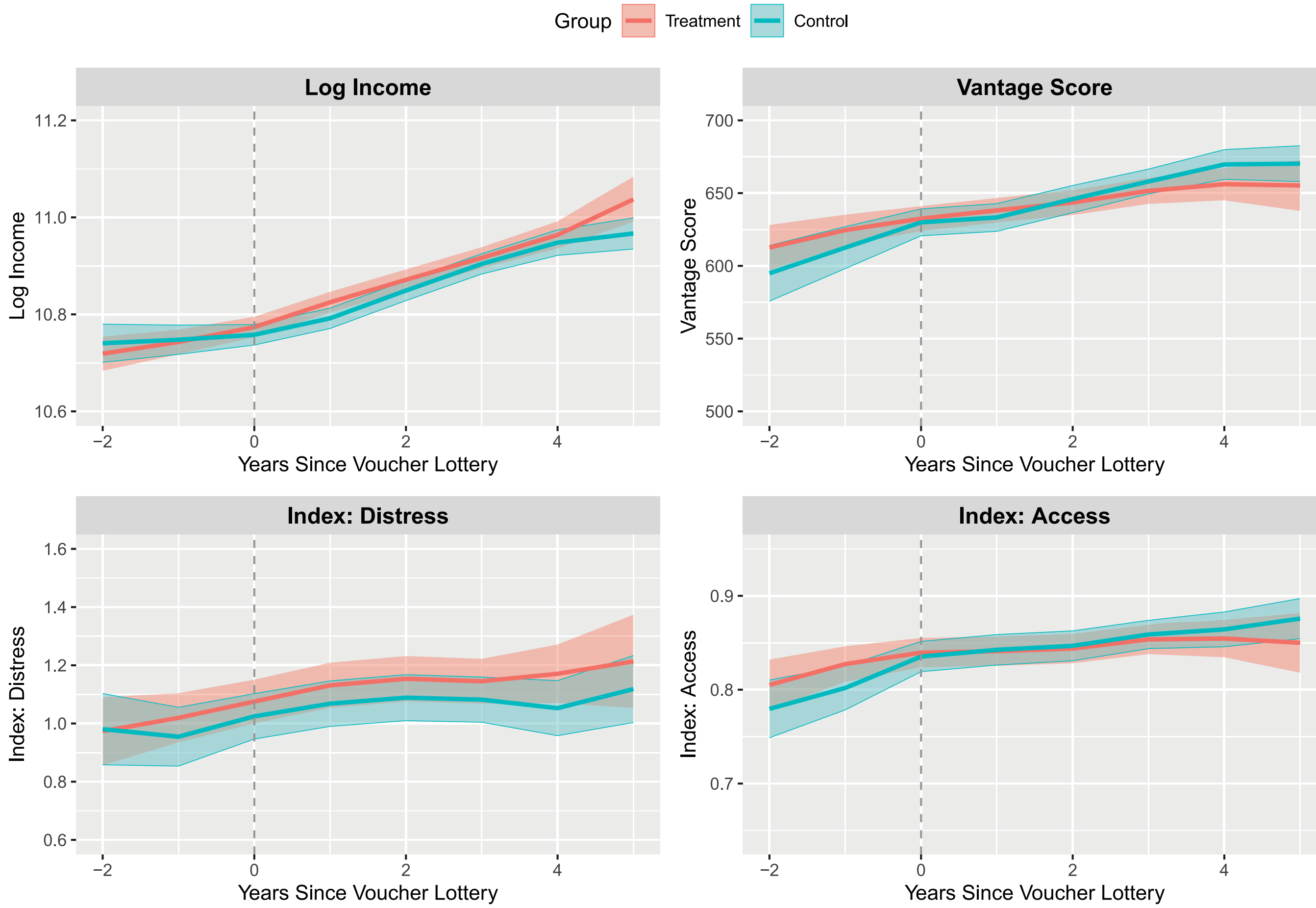

Fig. 1. Treatment effects on financial outcomes: Graphical evidence from credit bureau data. Outcome means for registrants in the treatment and control groups by year relative to the voucher lottery. The outcome variable is denoted in the title of each panel. The sample is restricted to registrants ($N = 2{,}081$) who are matched into the credit data for all six years. Shaded regions correspond to 95% CIs.

1.4.2.3. Dynamic effects. Next, we investigate whether the effects grow over time following Eq. **4**. The results are displayed in *SI Appendix*, Table S6. In contrast to ref. 11, we find no evidence that the effects grow over time. In fact, for income, the immediate effect (θ_1) is again close to zero with a point estimate of 0.3% (95% CI = −1.8%, 2.5%) and the interaction term (θ_2) is also close to zero, with a slightly negative point estimate at −0.1% (95% CI = −1.1%, 0.9%). When combining the coefficients, the model implies that the effect on income two years after the intervention is 0.0% with a 95% CI that rules out effects below −1.8% and above 1.8%. The implied effect four years after the intervention is a 0.2% drop, with a 95% CI that rules out effects below −3.4% and above 2.9%. The results for the other outcomes variables are similarly small and indistinguishable from zero.

Taken together, these results show that the voucher intervention considerably increased the uptake of citizenship but this did not lead to discernible improvements in financial outcomes as measured three years after the lottery.

1.4.3. Treatment effect estimates: Survey data.

1.4.3.1. Noneconomic integration. The null findings on economic returns raise the question of whether access to citizenship may have resulted in gains in other dimensions of immigrant integration. Previous observational research has identified positive impacts of access to citizenship on a variety of noneconomic integration outcomes, such as educational strategies of parents (61); social integration (16, 38); or political integration (15, 62).

To examine this question, we leverage the IPL-12 multidimensional integration measure (20). The findings for impacts on noneconomic integration are presented in Fig. 2. All outcomes are measured three years after the voucher lottery. Blue (red) dots and horizontal lines correspond to ITT (LATE) point estimates and CIs. Overall Integration shows the results for the overall IPL-12 integration index that aggregates all six dimensions (economic, political, social, psychological, navigational, and linguistic) into a single metric. Overall Integration shows the results when we omit the economic dimension and aggregate the remaining five dimensions. The rest show the results for each integration dimension separately.

We find no discernible evidence that access to citizenship improves any of the measured dimension of immigrant integration. The point estimates are close to zero and are not statistically significant at conventional levels. For example, the point estimate of the ITT effect on the overall integration index is a 0.0004 decrease with a 95% CI that rules out decreases larger than −0.015 and increases larger than 0.014. Given that the mean value of the integration index is 0.65 with a SD of 0.15, these estimates of the null effects are meaningfully precise.

1.4.3.2. Fears of deportation. Previous work (55) suggests that a high proportion of Latinos report deportation fears, even among lawful permanent residents, and that these reported fears are consistently high since the mid 2000s. This raises the question whether naturalization leads to a reduction in fears of deportation, given that citizenship provides immigrants the strongest legal protection against potential deportation. To examine this we use a survey question embedded in the 2021 follow up survey that asked registrants how much, if at all, they worry that they could be deported.[‡]

The results are shown in Fig. 3. We find that citizenship consistently reduced fears of deportation. For the pooled sample, the ITT estimates show a 0.15 point reduction in fears of deportation (95% CI from −0.24 to −0.06) measured on the four-point Likert scale that ranged from "worry a lot" (4) to "not at all" (1). Using the dichotomized measure there was a 7.4 percentage point drop (95% CI from −0.11 to −0.04) in the probability of having a high fear of deportation (answer options "worry a lot" or "some"). The LATE estimates suggest that citizenship

[‡]This outcome was not prespecified.

Table 2. Treatment effects: Three-year financial outcomes from credit bureau data

Panel A: Intent To treat effects (ITT)

	Income	Credit score	Financial distress		
	Log income	VantageScore	Delinquency	Collection	Index (distress)
Voucher	−0.000	−12.154*	0.104	0.025	0.065
	(0.012)	(5.019)	(0.063)	(0.031)	(0.041)
N	2,192	2,329	2,329	2,329	2,329
$\bar{Y}$	10.909	655.640	1.781	0.397	1.089
	Access to credit				
	Open line	Thick file	Has VantageScore	Index (access)	Credit line
Voucher	−0.029*	0.006	−0.009	−0.012	−0.023
	(0.012)	(0.017)	(0.006)	(0.009)	(0.051)
N	2,329	2,329	2,329	2,329	1,798
$\bar{Y}$	0.895	0.711	0.974	0.860	9.352

Panel B: Local average treatment effects (LATE)

	Income	Credit score	Financial distress		
	Log income	VantageScore	Delinquency	Collection	Index (distress)
Citizenship	0.025	−37.492*	0.139	0.099	0.122
	(0.041)	(17.472)	(0.216)	(0.101)	(0.138)
N	1,454	1,541	1,541	1,541	1,541
$\bar{Y}$	10.896	657.625	1.750	0.352	1.051
	Access to credit				
	Open line	Thick file	Has VantageScore	Index (access)	Credit line
Citizenship	−0.063	0.028	−0.025	−0.024	0.016
	(0.040)	(0.058)	(0.022)	(0.030)	(0.163)
N	1,541	1,541	1,541	1,541	1217
$\bar{Y}$	0.904	0.741	0.974	0.873	9.406

Notes: The outcome is denoted in the column header and is measured three years after voucher assignment. Panel A presents the ITT effects and Panel B shows the LATE effects. All regressions control for randomization block dummies, HH income, English language as well as the outcome value at baseline. Robust SEs are shown in parenthesis. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$.

reduced fears by 0.42 on the four point scale (95% CI from −0.66 to −0.18), and by 20.1 percentage points (95% CI from −0.30 to −0.10) for the binary measure, respectively. The other models in Fig. 3 show that this reduction in fear of deportation is stable regardless of whether the outcome was measured three, four, or five years after the lottery.

1.4.3.3. Mechanisms. Citizenship may enhance immigrants' economic integration by signaling long-term commitment that encourages investment in host-country human capital and by opening access to better employment opportunities (e.g., refs. 11 and 19. We examine both hypotheses. To test the first mechanism, we use credit bureau data on the likelihood of holding at least one student loan.[§] For the second, we analyze employment status, labor force participation, and occupational upgrading from follow-up surveys. The results are shown in *SI Appendix*, Table S8. We find no discernible evidence that access to citizenship improves human capital investments or other labor market outcomes.

1.4.4. Robustness checks. In *SI Appendix*, we report a rich set of robustness checks on the results we present here. These include quantile regression models, randomization inference, various alternative weighting procedures, adjustments for multiple testing and sample attrition, and cohort-based subgroup analysis. Our null results are consistent across all alternative model specifications, methodologies, and samples. We find no evidence that citizenship leads to improved integration outcomes.

1.4.5. Selection bias. Our experiment offers a unique opportunity to compare the effect estimates from a standard observational regression versus those from the randomized experiment. This exercise might help reconcile the tension between our null findings with the observational literature on citizenship that has tended to find positive economic effects of access to citizenship. A key concern for the nonexperimental regression is that immigrants self-select into citizenship based on unobserved confounders such as ability and motivation (10, 11, 17, 19, 63). In other words, those who choose to apply for citizenship might have better outcomes even in absence of naturalization.

To test for selection bias, we regressed the three-year outcomes on the citizenship indicator and the baseline covariates, including only respondents in the control group. *SI Appendix*, Table S11 presents the results. Immigrants who select into citizenship indeed have better three-year financial outcomes, with most outcome variables being statistically significant at conventional levels. We find that citizenship is associated with gains in the VantageScore (26.2 points, P-value $= 0.01$), probability of having an open line (5.6%, $P = 0.004$), and a thick file (9.5%, $P = 0.004$), and reductions in delinquencies (−0.31, $P = 0.032$), reduction in collections (−0.16 , $P = 0.008$), and financial distress index (0.24 units, $P = 0.011$), and others.[¶]

1.4.6. External validity. We expect our experiment to have external validity for the U.S. context. First, it was conducted in New York, a major urban immigrant labor market similar to other large U.S. metro areas with high immigrant populations, such as Los Angeles, Chicago, Houston, and Atlanta. Second, the experiment was embedded in a program run by nonprofits that commonly support low-income immigrants pursuing naturalization. This gave assurance that our experiment was not perceived as unusual, or a special intervention. Third,

[§]This analysis was not preregistered.

[¶]These tests for selection bias are likely conservative for three reasons. First, we reduce statistical power by only using the control group. In fact, we do find a statistically significant effect on income when using the entire sample instead of just the control group ($P = 0.05$). We also find similar significant effects on income when using the panel model specification only with the control group ($P = 0.06$) or with the full sample ($P = 0.004$). In fact, when we run the dynamic version of the panel regression ignoring the experimental design, we reproduce the results in Bratsberg et al. (11) that the effect of citizenship on income grows over time; the interaction term between citizenship and years since citizenship has a P-value of 0.044 in the control group only and 0.039 in the full sample. According to these models citizenship would lead to about a 7% income gain within four years. Second, our sample consists of only eligible immigrants who were motivated enough to register for a naturalization program. Third, our data allows us to control for pretreatment outcomes which further decreases the selection bias.

Table 3. Treatment effects: Panel estimates for financial outcomes (credit bureau data)

Panel A: Intent to treat effects (ITT)

	Income	Credit score	Financial distress		
	Log income	VantageScore	Delinquency	Collection	Index (distress)
Voucher	0.002	−6.754	0.009	0.022	0.016
	(0.009)	(4.609)	(0.050)	(0.025)	(0.032)
N	11,949	12,486	12,486	12,486	12,486
$\bar{Y}$	10.842	638.841	1.732	0.420	1.076

	Access to credit				
	Open line	Thick file	Has VantageScore	Index (access)	Credit line
Voucher	−0.010	−0.015	−0.010	−0.012	−0.057
	(0.010)	(0.015)	(0.006)	(0.008)	(0.041)
N	12,486	12,486	12,486	12,486	10,324
$\bar{Y}$	0.877	0.674	0.967	0.839	9.072

Panel B: Local average treatment effects (LATE)

	Income	Credit score	Financial distress		
	Log income	VantageScore	Delinquency	Collection	Index (distress)
Citizenship	0.018	−11.929	−0.046	0.059	0.006
	(0.018)	(9.571)	(0.102)	(0.053)	(0.067)
N	7,819	8,184	8,184	8,184	8,184
$\bar{Y}$	10.828	641.762	1.684	0.404	1.044

	Access to credit				
	Open line	Thick file	Has VantageScore	Index (access)	Credit line
Citizenship	0.008	−0.004	−0.019	−0.005	−0.045
	(0.021)	(0.031)	(0.014)	(0.016)	(0.080)
N	8,184	8,184	8,184	8,184	6,813
$\bar{Y}$	0.884	0.690	0.967	0.847	9.132

Notes: Each entry is an estimated coefficient from a separate regression of the full panel model. Panel A presents the ITT effects and Panel B shows the LATE effects. All regressions control for individual and year fixed effects. The sample is restricted to individuals who are matched in the balanced six-year panel. SEs are clustered at the individual level and shown in parenthesis. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$.

our sample consisted of low-income immigrants with permanent residency–the group for which prior research (e.g., refs. 11 and 19) finds the highest returns to citizenship. Fourth, our null effects were consistent across various subgroups and a wide range of economic integration outcomes, suggesting limited discernible effect heterogeneity. Fifth, the compliers are similar to the overall sample across the covariates in both the registration and the credit bureau data (*SI appendix*).

To formally assess external validity, we combine our experimental data with population-level data from two target groups: 1) all low-income, naturalization-eligible immigrants in New York, and 2) those in all U.S. metro areas. The population data come from the 2015 to 2019 5-y American Community Survey (ACS) (64).[#] The ACS includes many of the same pretreatment covariates as our experimental data: country of origin, age, gender, marital status, education, and pretreatment income. We estimate population average ITT effects for both target groups using the three-year outcome models and three sets of methods (65–67)–weighting to balance covariates across the experimental and target populations, flexible outcome modeling to relax OLS's linearity assumption, and doubly robust estimators.

SI Appendix, Fig. S4 reports population average effects on logged household income and the IPL-12 integration index. We find consistent null effects across both target populations and all estimators, matching results in the experimental sample. These findings support the external validity of our experiment and reinforce the limited treatment effect heterogeneity observed.

[#]We restrict the ACS sample to foreign-born, noncitizens aged 18+, in the United States for at least five years, with household income below 300% of the FPG.

2. Discussion

The results from this experimental study of the effects of citizenship provide a somewhat sobering picture of the impact of access to naturalization on the economic prospects of immigrants. While the voucher intervention resulted in a large increase in citizenship uptake, it did not translate into discernible improvements in short to medium term financial outcomes. These results are in contrast with the most highly cited research on the economic impact of citizenship in the United States which found considerable gains from citizenship over permanent residency (11–13). The results are also in contrast with the conventional wisdom in the literature that has documented positive effects of citizenship in various other countries (19) and summarized in the Appendix. Similarly, we found no discernible effects on related outcomes such as financial distress, access to credit, and credit scores. There is also no support for the idea that access to citizenship led to higher educational investment, employment, labor force participation, or occupational upgrading. Moreover, we found no discernible effects on noneconomic integration using an index that captures social, political, psychological, navigational, and linguistic integration. Last, we found that citizenship had a lasting impact on reducing fears of deportation.

How to reconcile these differences in the estimated effects? One possible interpretation is that the previous observational

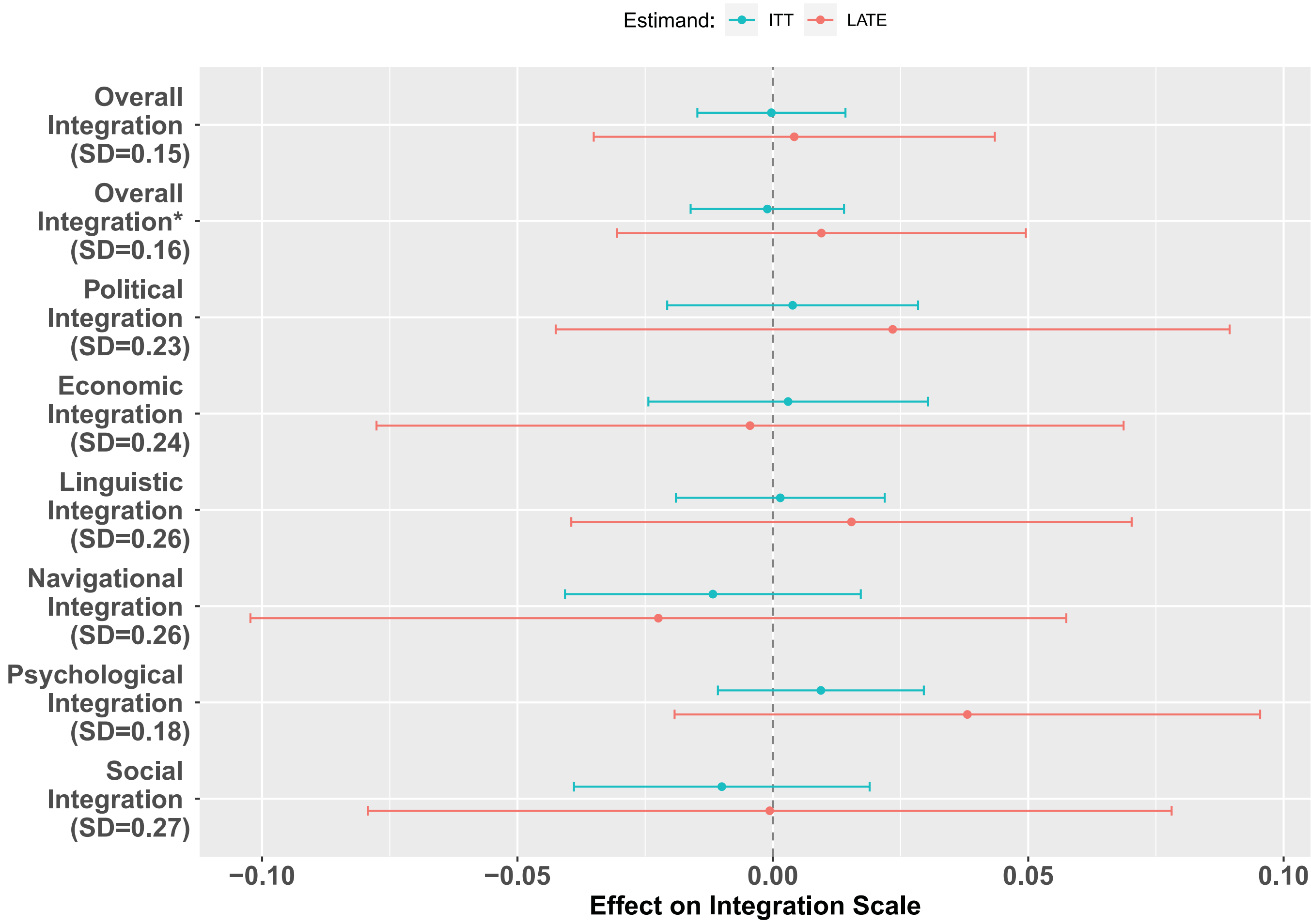

Fig. 2. Treatment effects on noneconomic outcomes: Graphical evidence from survey data. Treatment effect estimates (ITT in blue, LATE in red) of citizenship on noneconomic integration. The outcome variable is denoted in the left bar. The sample is restricted to registrants who responded to the follow-up surveys. $N = 1{,}184$ for Overall Integration ITT estimate. Horizontal bars correspond to 95% CIs.

estimates were driven by positive selection. Indeed, when we ignore the experimental variation and compare naturalized with nonnaturalized immigrants, we document the presence of positive selection effects with a significant citizenship premium. These effects are present despite the fact that our sample was limited to only motivated immigrants who registered for the program. It stands to reason that these selection effects would be even stronger in observational designs that cannot control for motivation.

Taken together, our results cast some doubt on the prominent claim in the literature that facilitating access to citizenship on top of permanent residency provides an effective policy lever to accelerate the economic integration of immigrants (11, 12, 17, 37). At least within the context of this randomized experiment on access to citizenship, we do not find discernible evidence of an economic citizenship premium in the short to medium term.

While our analysis suggests that our experiment has external validity for the current U.S. context, it is important to emphasize that our results do not rule out the possibility of an economic citizenship premium in other contexts, esp. other countries or time periods. Potential moderating factors include labor market flexibility, the intensity of labor market discrimination based on citizenship status, and the difficulty of obtaining citizenship. For example, ref. 17 documents positive long-term economic effects of citizenship in Switzerland, a country with more demanding requirements for naturalization compared to the United States. It is possible that, in a setting where naturalization requires a more costly investment, the signaling value of citizenship is higher, giving rise to a premium. Another factor may be historical time; in earlier eras, for example during America's Great Migration, a quasi-experimental study (23) revealed a substantial citizenship premium. More experimental research from other contexts is needed to better evaluate the extent to which citizenship can act as a catalyst for integration or have an effect on other outcomes. Similarly, our results are limited to short to medium term outcomes, and follow-up work on our sample (or other experimental samples) is needed to consider the impacts on longer term outcomes.

Last, recall that our results concern the impact of access to citizenship among immigrants who have already attained permanent residency in the United States. These results therefore do not speak to whether there may be significant economic returns to obtaining other immigrant statuses such as permanent residency (68, 69) or other forms of legal protection such as deferred action (70, 71). Moreover, the results do not show that citizenship has no value for immigrants. In fact, naturalization puts immigrants on a virtually equal legal footing with natives and opens the door to benefits such as the ability to travel and return on a U.S. passport, access to restricted jobs, the right to vote, protection from deportation, and the opportunity to sponsor family members for visas. Consistent with this we did find psychological benefits in terms of reduced fears of deportation.

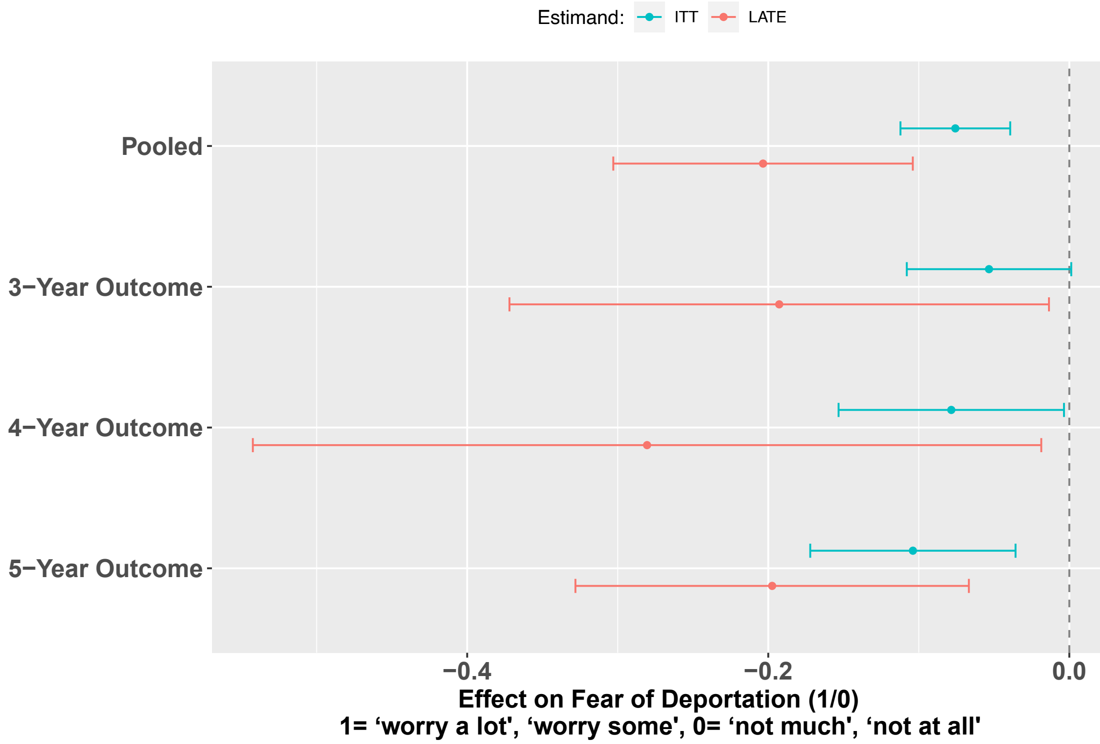

Fig. 3. Treatment effects on fear of deportation: Graphical evidence from survey data. Treatment effect estimates (ITT in blue, LATE in red) of citizenship on deportation fears. The sample is denoted in the left bar. The sample is restricted to registrants who responded to the follow-up surveys. $N = 1{,}290$ for Pooled ITT estimate. Horizontal bars correspond to 95% CIs.

Data, Materials, and Software Availability. Some study data are available: A coarsened version of the survey data is available upon request and subject to a data use agreement. The credit bureau data cannot be shared with a third party, per the contract governing its use. The data analysis code is available at Zenodo (72).

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1. WB OECD, ILO, IMF, "Towards a framework for fair and effective integration of migrants into the labour market" (Tech. Rep., 2016).
2. R. Abramitzky, L. Boustan, Immigration in American economic history. *J. Econ. Lit.* **55**, 1311–1345 (2017).
3. B. R. Chiswick, The effect of Americanization on the earnings of foreign-born men. *J. Polit. Econ.* **86**, 897–921 (1978).
4. W. McManus, W. Gould, F. Welch, Earnings of Hispanic men: The role of English language proficiency. *J. Law Econ.* **1**, 101–130 (1983).
5. G. G. Cain, The economic analysis of labor market discrimination: A survey. *Handb. Labor Econ.* **1**, 693–785 (1986).
6. G. J. Borjas, Assimilation and changes in cohort quality revisited: What happened to immigrant earnings in the 1980s? *J. Law Econ.* **13**, 201–245 (1995).
7. R. M. Friedberg, You can't take it with you? Immigrant assimilation and the portability of human capital. *J. Law Econ.* **18**, 221–251 (2000).
8. P. A. Riach, J. Rich, Field experiments of discrimination in the market place. *Econ. J.* **112**, F480–F518 (2002).
9. P. Oreopoulos, Why do skilled immigrants struggle in the labor market? A field experiment with thirteen thousand resumes. *Am. Econ. J. Econ. Pol.* **3**, 148–171 (2011).
10. National Academies of Sciences, Engineering, and Medicine, *The Integration of Immigrants into American Society*, M. C. Waters, M. G. Pineau, Eds. (The National Academies Press, Washington, DC, 2015).
11. B. Bratsberg, J. F. Ragan Jr., Z. M. Nasir, The effect of naturalization on wage growth: A panel study of young male immigrants. *J. Law Econ.* **20**, 568–597 (2002).
12. P. Miller, B. Chiswick, "Language in the immigrant labour market" in *Immigration, Language & Ethnicity: Canada and the United States*, B. R. Chiswick, Ed. (American Enterprise Institute, 1992), pp. 229–296.
13. F. Mazzolari, Dual citizenship rights: Do they make more and richer citizens? *Demography* **46**, 169–191 (2009).
14. Y. Govind, Is naturalization a passport for better labor market integration? Evidence from a quasi-experimental setting. (2021).
15. J. Hainmueller, D. Hangartner, G. Pietrantuono, Naturalization fosters the long-term political integration of immigrants. *Proc. Natl. Acad. Sci. U.S.A.* **112**, 12651–12656 (2015).
16. J. Hainmueller, D. Hangartner, G. Pietrantuono, Catalyst or crown: Does naturalization promote the long-term social integration of immigrants? *Am. Polit. Sci. Rev.* **111**, 256–276 (2017).
17. J. Hainmueller, D. Hangartner, D. Ward, The effect of citizenship on the long-term earnings of marginalized immigrants: Quasi-experimental evidence from Switzerland. *Sci. Adv.* **5**, eaay1610 (2019).
18. J. Hainmueller *et al.*, A randomized controlled design reveals barriers to citizenship for low-income immigrants. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 939–944 (2018).
19. T. Liebig, "Main findings of the joint EC/OECD seminar on naturalisation and the socio-economic integration of immigrants and their children" in *Naturalisation: A Passport for the Better Integration of Immigrants?* OECD, Ed. (OECD Publishing, 2011), pp. 15–20.
20. N. Harder *et al.*, Multidimensional measure of immigrant integration. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 11483–11488 (2018).
21. B. Bratsberg, J. F. Ragan Jr., Z. M. Nasir, The effect of naturalization on wage growth: A panel study of young male immigrants. *J. Law Econ.* **20**, 568–597 (1992).
22. M. Sumption, S. Flamm, "The economic value of citizenship for immigrants in the United States" (Migration Policy Institute, Washington, DC, 2012).
23. P. Catron, The citizenship advantage: Immigrant socioeconomic attainment in the age of mass migration. *Am. J. Sociol.* **124**, 999–1042 (2019).
24. F. Mazzolari, *Determinants and effects of naturalization. The role of dual citizenship laws* (Rutgers University, Working paper, 2006).
25. M. Pastor, J. Scoggins, "Citizen gain: The economic benefits of naturalization for immigrants and the economy" (Tech. Rep., Equity Research Institute, 2012).
26. G. Picot, F. Hou, "Citizenship acquisition in Canada and the United States: Determinants and economic benefit" in *Naturalisation: A Passport for the Better Integration of Immigrants?*, T. Liebig, Ed. (OECD Publishing, 2011), pp. 154–183.
27. H. Shierholz, "The effects of citizenship on family income and poverty" (Tech. Rep. 256, Economic Policy Institute, 2010).

28. H. Zhou, S. Lee, Effects of US citizenship on wages of Asian immigrant women. *Int. J. Soc. Welf.* **22**, 420–430 (2013).
29. M. E. Enchautegui, L. Giannarelli, "The economic impact of naturalization on immigrants and cities" (Tech. Rep., Urban Institute, 2015).
30. S. Pivnenko, D. DeVoretz, The recent economic performance of Ukrainian immigrants in Canada and the U.S. (Discussion Paper No. 913, 2003). https://ssrn.com/abstract=464661.
31. D. DeVoretz, S. Pivnenko, The economic causes and consequences of Canadian citizenship. *Int. Migr. Integr.* **6**, 435–468 (2005).
32. R. Pendakur, P. Bevelander, Citizenship, enclaves and earnings: Comparing two cool countries. *Citizsh. Stud.* **18**, 384–407 (2014).
33. F. Mata, "Patterns of acquiring citizenship" in *Immigrant Canada: Demographic, Economic and Social Challenges*, S. S. Halli, L. Driedger, Eds. (University of Toronto Press, 1999), pp. 163–182.
34. M. F. Steinhardt, Does citizenship matter? The economic impact of naturalizations in Germany. *Labour Econ.* **19**, 813–823 (2012).
35. A. Constant, L. Gataullina, K. F. Zimmermann, "Naturalization proclivities, ethnicity and integration" (DIW Discussion Papers No. 755, German Institute for Economic Research, 2007).
36. R. Euwals, J. Dagevos, M. Gijsberts, H. Roodenburg, Citizenship and labor market position: Turkish immigrants in Germany and the Netherlands. *Int. Migr. Rev.* **44**, 513–538 (2010).
37. C. Gathmann, N. Keller, Access to citizenship and the economic assimilation of immigrants. *Econ. J.* **128**, 3141–3181 (2018).
38. C. Gathmann, O. Monscheuer, Naturalization and citizenship: Who benefits? *IZA World Labor* **125**, e223607 (2020).
39. C. Sajons, Birthright citizenship and parental labor market integration. *Labour Econ.* **57**, 1–22 (2019).
40. K. Scott, "The economics of citizenship: Is there a naturalization effect?" in *The Economics of Citizenship*, P. Bevelander, D. DeVoretz, Eds. (Malmo University Press, 2008), pp. 107–126.
41. M. Engdahl, "The impact of naturalisation on labour market outcomes in Sweden" in *Naturalisation: A Passport for the Better Integration of Immigrants?*, T. Liebig, Ed. (OECD Publishing, 2011), pp. 99–130.
42. P. Bevelander, *Immigrant Employment Integration and Structural Change in Sweden* (Almqvist and Wiksell International, 2000), pp. 1970–1995.
43. I. Kogan, Ex-yugoslavs in the Austrian and Swedish labour markets: The significance of the period of migration and the effect of citizenship acquisition. *J. Ethnic Migr. Stud.* **29**, 595–622 (2010).
44. J. Helgertz, P. Bevelander, A. Tegunimataka, Naturalization and earnings: A Denmark-Sweden comparison. *Eur. J. Popul.* **30**, 337–359 (2014).
45. B. Bratsberg, O. Raaum, "The labor market outcomes of naturalised citizens in Norway" in *Naturalisation: A Passport for the Better Integration of Immigrants?*, T. Liebig, Ed. (OECD Publishing, 2011), pp. 184–205.
46. J. E. Hayfron, "The economics of Norwegian citizenship" in *The Economics of Citizenship*, P. Bevelander, D. DeVoretz, Eds. (Malmo University, 2008), pp. 89–104.
47. M. F. Steinhardt, "The impact of naturalisation on immigrant labour market integration in Germany and Switzerland" in *Naturalisation: A Passport for the Better Integration of Immigrants?*, T. Liebig, Ed. (OECD Publishing, 2011), pp. 146–153.
48. F. Peters, M. Vink, H. Schmeets, Anticipating the citizenship premium: Before and after effects of immigrant naturalisation on employment. *J. Ethn. Migr. Stud.* **44**, 1051–1080 (2017).
49. F. Peters, H. Schmeets, M. Vink, Naturalisation and immigrant earnings: Why and to whom citizenship matters. *Eur. J. Popul.* **36**, 511–545 (2019).
50. P. Bevelander, J. Veenman, Naturalization and employment integration of Turkish and Moroccan immigrants in the Netherlands. *J. Int. Migr. Integr.* **7**, 327–349 (2006).
51. D. Fougère, M. Safi, Naturalization and employment of immigrants in France (1968–1999). *Int. J. Manpow.* **30**, 83–96 (2009).
52. J. Jarreau, "Naturalization policy and the economic integration of immigrants: New evidence for france" (Tech. Rep., Paris Dauphine University, 2020).
53. V. Corluy, I. Marx, G. Verbist, Employment chances and changes of immigrants in Belgium: The impact of citizenship. *Int. J. Comp. Appl. Crim. Just.* **52**, 350–368 (2011).
54. OECD, *Naturalisation: A Passport for the Better Integration of Immigrants?* (OECD, 2011).
55. A. L. Asad, Latinos' deportation fears by citizenship and legal status, 2007 to 2018. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 8836–8844 (2020).
56. P. W. Holland, Causal inference, path analysis and recursive structural equations models. *ETS Res. Rep. Ser.* **1988**, 1–50 (1988).
57. G. W. Imbens, J. D. Angrist, Identification and estimation of local average treatment effects. *Econometrica J. Econometric Soc.* **62**, 467–475 (1994).
58. E. Duflo, R. Glennerster, M. Kremer, Using randomization in development economics research: A toolkit. *Handb. Dev. Econ.* **4**, 3895–3962 (2007).
59. H. S. Bloom, "The core analytics of randomized experiments for social research" in *The SAGE Handbook of Social Research Methods*, P. Alasuutari, L. Bickman, J. Brannen, Eds. (Sage Publications, 2008), pp. 115–133.
60. A. Abadie, S. Athey, G. W. Imbens, J. M. Wooldridge, When should you adjust standard errors for clustering? *Q. J. Econ.* **138**, 1–35 (2023).
61. C. Felfe, H. Rainer, J. Saurer, Why birthright citizenship matters for immigrant children: Short-and long-run impacts on educational integration. *J. Law Econ.* **38**, 143–182 (2020).
62. A. Street, The political effects of immigrant naturalization. *Int. Migr. Rev.* **51**, 323–343 (2017).
63. M. Engdahl, "Naturalizations and the economic and social integration of immigrants" (Tech. Rep., Working Paper No. 2014:11 2014).
64. S. Ruggles *et al.*, Integrated public use microdata series: Version 9.0 [dataset]. *Minneap. Univ. Minn.* **23**, 56 (2019).
65. N. Egami, E. Hartman, Elements of external validity: Framework, design, and analysis. *Am. Polit. Sci. Rev.* **117**, 1070–1088 (2023).
66. E. Hartman, R. Grieve, R. Ramsahai, J. S. Sekhon, From sate to patt: Combining experimental with observational studies to estimate population treatment effects. *JR Stat. Soc. Ser. A Stat. Soc.* **10**, 1111 (2015).
67. C. Blattman, N. Fiala, S. Martinez, Generating skilled self-employment in developing countries: Experimental evidence from Uganda. *Q. J. Econ.* **129**, 697–752 (2014).
68. J. A. Phillips, D. S. Massey, The new labor market: Immigrants and wages after IRCA. *Demography* **36**, 233–246 (1999).
69. E. U. Cascio, E. G. Lewis, Distributing the green (cards): Permanent residency and personal income taxes after the immigration reform and control act of 1986. *J. Public Econ.* **172**, 135–150 (2019).
70. J. Hainmueller *et al.*, Protecting unauthorized immigrant mothers improves their children's mental health. *Science* **357**, 1041–1044 (2017).
71. N. G. Pope, The effects of dacamentation: The impact of deferred action for childhood arrivals on unauthorized immigrants. *J. Public Econ.* **143**, 98–114 (2016).
72. J. Hainmueller, Replication Code: The socioeconomic returns to citizenship: A randomized controlled trial. Zenodo. https://doi.org/10.5281/zenodo.18202916. Deposited 9 January 2026.